

CODEALPHA

TASK 2

E-Commerce Sales Data Analysis

Exploratory Data Analysis (EDA) Report

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Dataset	CodeAlpha_ECommerce_Sales_Dataset.xlsx
Analysis Tool	Python, Pandas, Jupyter Notebook
Task	Exploratory Data Analysis (EDA)

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1. Introduction

This project presents an Exploratory Data Analysis (EDA) of an e-commerce sales dataset. The purpose of the analysis is to understand the structure of the data, identify data-quality issues, examine sales and revenue trends, discover patterns across categories, cities and products, and identify unusual revenue observations.

2. Objectives of the Analysis

- Understand the dataset structure, variables and data types.
- Identify missing values and handle them appropriately.
- Analyze revenue trends across months and business dimensions.
- Compare revenue across product categories, cities and products.
- Identify statistical outliers in revenue.
- Verify that recorded revenue is consistent with quantity, price and discount.
- Summarize the main findings and insights for further visualization and business analysis.

3. Dataset Overview

The loaded dataset contains 1,500 sales records and 12 columns. The variables cover order identification, date, location, product information, quantity, pricing, discount, revenue and time-period fields.

Metric	Result
Total records	1,500
Total columns	12
Total quantity sold	3,714
Total revenue	13,983,089.70
Average revenue per record	9,322.06

Columns used in the analysis: Order_ID, Order_Date, City, Category, Product, Quantity, Unit_Price, Discount, Revenue, Month, Month_Number and Quarter.

4. Data Cleaning and Quality Checks

4.1 Missing Values

The initial missing-value check showed missing observations only in the City and Product columns. There were 5 missing City values and 5 missing Product values. The remaining columns had no missing values.

Column	Missing Values	Action
City	5	Filled with "Unknown"
Product	5	Filled with "Unknown"
Other columns	0	No action required

4.2 Revenue Consistency Check

Revenue was independently recalculated using the relationship:

$$\text{Calculated Revenue} = \text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount})$$

The maximum absolute difference between the recorded Revenue and the calculated Revenue was approximately 7.28×10^{-12} , which is effectively zero for this analysis. This indicates that the revenue values are consistent with the quantity, unit price and discount fields.

4.3 Duplicate and Descriptive Checks

Duplicate checks and descriptive statistics were used as part of the EDA workflow. The numerical fields were summarized using count, mean, standard deviation, minimum, quartiles and maximum.

Variable	Mean	Minimum	Maximum
Quantity	2.476	1	4
Unit Price	4,092.70	151	17,962
Discount	0.08	0	0.20
Revenue	9,322.06	155.80	70,820.00

5. Exploratory Data Analysis

5.1 Monthly Revenue Trend

Monthly revenue was grouped by month number and month name to preserve the chronological order from January to December.

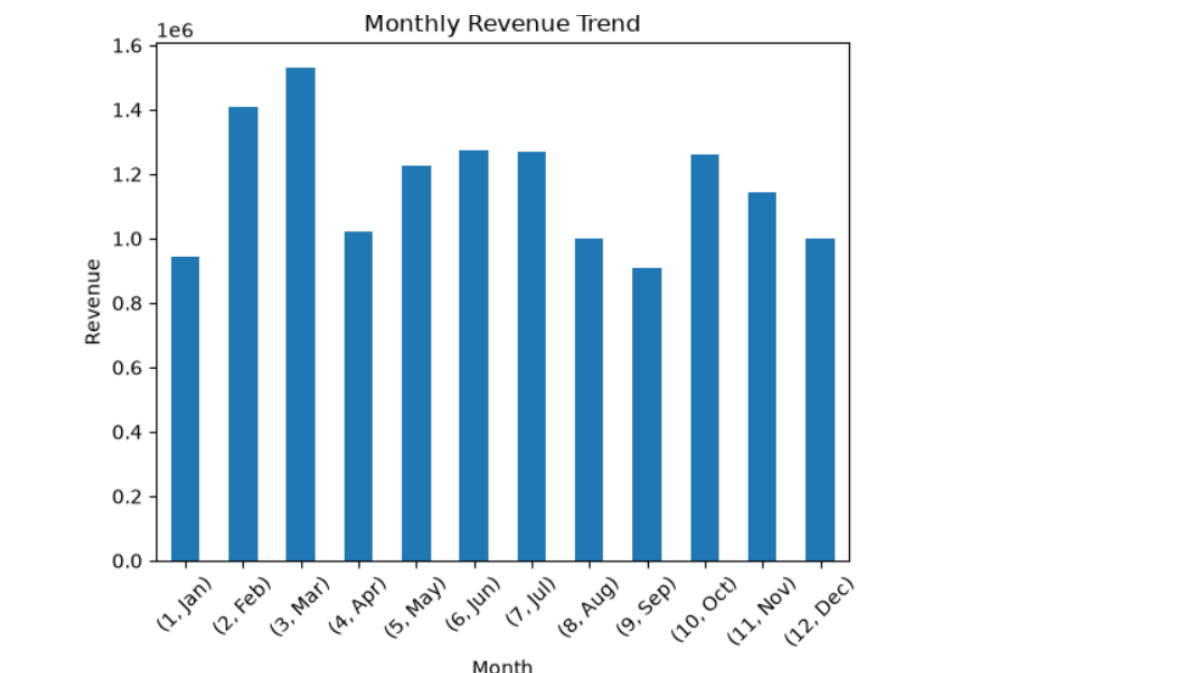


Figure 1: Monthly Revenue Trend

Observation: Revenue varies noticeably across the year. March recorded the highest monthly revenue (₹15,30,403.15), while September recorded the lowest monthly revenue (₹9,10,548.90). The pattern indicates that monthly revenue was not uniform and experienced several increases and decreases.

5.2 Category-wise Revenue

Revenue was aggregated by Category to compare the contribution of different product categories. Electronics generated the highest category-level revenue, amounting to ₹89,37,601.40.

5.3 City-wise Revenue

Revenue was aggregated by City to examine geographic performance. Mumbai recorded the highest city-level revenue in the analysis, with ₹16,08,805.10.

5.4 Product-wise Revenue

The products were grouped by total revenue and the top products were examined. Power Bank recorded the highest product-level revenue, with ₹21,21,624.25.

5.5 Quantity and Quarter Analysis

Revenue was also grouped by Quantity and Quarter to examine how transaction size and broader time periods relate to revenue. These comparisons were visualized in the Jupyter Notebook as part of the EDA workflow.

6. Outlier / Anomaly Analysis

The Interquartile Range (IQR) method was used to identify unusually high or low Revenue observations.

$$\text{IQR} = Q3 - Q1$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Outlier Metric	Result
Number of Revenue outliers	131
Upper Revenue threshold	26,298.94 approximately
Maximum observed Revenue	70,820.00

The 131 observations were flagged as statistical outliers. They were not automatically deleted because a statistical outlier is not necessarily an erroneous transaction. For example, high-value orders can be legitimate business transactions. The outlier flag should therefore be treated as a point for further investigation.

7. Key Findings and Insights

- The dataset contains 1,500 sales records across 12 variables.
- A total quantity of 3,714 units was recorded, with total revenue of ₹1,39,83,089.70.
- Electronics contributed the highest category-level revenue at ₹89,37,601.40.
- Mumbai recorded the highest city-level revenue at ₹16,08,805.10.
- Power Bank generated the highest product-level revenue at ₹21,21,624.25.
- March recorded the highest monthly revenue at ₹15,30,403.15.
- September recorded the lowest monthly revenue at ₹9,10,548.90.
- Five City values and five Product values were missing and were handled as Unknown.
- Revenue was internally consistent with Quantity, Unit Price and Discount based on the calculation check.
- The IQR method identified 131 revenue outliers for further investigation.

8. Conclusion

The exploratory analysis provided a structured view of the e-commerce sales dataset. The analysis covered data structure, missing-value handling, descriptive statistics, monthly trends, category and city performance, product-level revenue, quantity and quarter comparisons, and revenue outliers. The results provide a useful foundation for the next stage of the project, where the findings can be presented through more detailed charts, dashboards and business-oriented visualizations.

9. Tools and Methodology

Component	Used
Programming Language	Python
Data Analysis Library	Pandas
Excel Reading	OpenPyXL
Visualization	Matplotlib
Environment	Jupyter Notebook / JupyterLab
Dataset Format	Excel (.xlsx)

End of Report